



Rivers beneath the surface: how riverine inputs shape sedimentation in a large peri-alpine lake

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Abstract

Climate change is increasing the frequency and intensity of floods, droughts, and storms, yet their effects on sedimentation and biogeochemical cycling in large lakes remain poorly constrained. Riverine inflows play a central role by controlling the vertical and horizontal transport of suspended matter through density-driven intrusions. Here, we quantify sediment flux dynamics over 5 years, using sediment traps and high-frequency environmental observations in Lake Geneva, a large peri-alpine lake influenced by the Rhône River. Fluxes of total mass, organic matter, calcite, and minerals reveal higher particle export below 30 m than at 10 m, particularly in summer when interflows intrude at the thermocline. Peak fluxes coincided with high river discharge, with minerals fractions dominating total export. Extreme events emerge as the primary driver of variability, producing distinct sedimentation responses. Floods produced the highest particle fluxes and promoted efficient transport to depth, whereas drought conditions strongly reduced sediment supply, with short-lived storm-driven pulses. In contrast, wind-driven storms drove sediment inputs through resuspension and redistribution. These results demonstrate that the interaction between fluvial inputs, stratification, and hydrodynamics governs sediment export in large, river-influenced lakes worldwide. This study provides a transferable framework for predicting sediment responses to hydroclimatic extremes in such systems.

Keywords Sediment · Sinking fluxes · River intrusion · Extreme events · Lake Geneva

Introduction

Suspended particulate matter (SPM) plays a crucial role in lake ecosystems, influencing biogeochemical cycles, water quality, and sedimentation processes. Comprising a diverse mix of particles varying in size, shape, and composition, SPM originates from both autochthonous sources, such as phytoplankton, zooplankton, and carbonate precipitates, and allochthonous sources, including riverine inputs, surface runoff, and atmospheric deposition. These particles are dynamic within the aquatic environment, undergoing aggregation, disaggregation, mineralization, and transformation

during their transport and settling through the water column (Jonsson et al. 2001; Fahrner et al. 2008; Deng et al. 2022; Ho et al. 2022).

Through sedimentation to the lake bed and mineralization within the water column and at the sediment–water interface, particulate matter regulates nutrient cycling and organic matter burial (Müller et al. 2005; Pasche et al. 2012). Sediment traps have been widely used to investigate vertical fluxes of SPM in lakes, providing insights into seasonal variability, particle composition, and sedimentation rates (Bloesch and Burns 1980; Sigg et al. 1987; Kirilova et al. 2008; Pasche et al. 2010; Stockhecke et al. 2012). In large lakes (depth > 40 m), these studies reveal that most organic matter mineralization occurs in the water column rather than at the sediment–water interface, with mineralization efficiency generally decreasing with depth (Schweifel et al. 2018; Fernández Castro et al. 2021b; Steinsberger et al. 2021). Labile organic material decomposes rapidly in upper layers, while more refractory carbon persists at greater depths (Sobek et al. 2009). Sediment focusing, driven by lateral transport and enhanced nearshore accumulation, further

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influences the vertical and spatial distribution of particles and associated contaminants (Graham et al. 2016).

Among the processes shaping sedimentation, river inflows are particularly important, as they regulate hydrodynamics, sedimentation, and the transport of particle-bound contaminants in lakes. The fate of river-derived particles depends on the density contrast between the inflowing water and the receiving lake. Depending on their relative density, river inflows can generate three distinct types of flows: underflows, where dense, sediment-laden water sinks along the bottom; interflows, which intrude at intermediate depths corresponding to neutral buoyancy; and overflows, where lighter inflows remain at the surface (Cortés et al. 2014a; Silva et al. 2019). Understanding these flow types is critical for predicting the vertical distribution of SPM, nutrient fluxes, and contaminant pathways in lacustrine systems (Goharrokhi et al. 2021; Khazaei et al. 2021; Zhu et al. 2024).

Examples from large lakes highlight the dominant role of river inflows in sediment dynamics. In Lake Michigan, intensified waves and rising water levels have amplified sediment suspension and coastal erosion, with transport patterns varying across the basin (Thorez et al. 2024; Zhu et al. 2024). In Lake Winnipeg, riverine inputs dominate sediment supply in some basins, while shoreline erosion is more important in others, reflecting the complex interplay of local and catchment-scale sources (Goharrokhi et al. 2021). In Lake Maggiore, river intrusions at intermediate depths drive sediment transport and deposition, with vertical confinement of flows enhancing particle settling (Scheu et al. 2015, 2018). Studies of deltaic systems further illustrate how river diversions control sediment retention and redistribution in receiving basins (Vonk et al. 2016; Allison et al. 2017). In Lake Onego, the Shuya River delivers mineral-rich suspended matter that dominates near-shore sediments and declines toward the open basin (Chmiel et al. 2020; Belkina et al. 2023). In Lake Constance, river-borne particulate organic matter contributes strongly to sedimentation, but its influence decreases with distance from major inflows and varies seasonally (Fuentes et al. 2013). Similarly, in Lake Brienz, sediment supplied by the Aare and Lütschine Rivers accumulates rapidly near the deltas and decreases toward the basin plain (Fabbri et al. 2021). However, these studies span a wide range of temporal scales, from short-term process-based observations lasting hours to months (e.g., Scheu et al. 2015; Allison et al. 2017; Thorez et al. 2024), to multiyear but low-resolution sedimentation records integrating fluxes over months to years (e.g., Belkina et al. 2023), and to longer-term reconstructions that lack event-scale resolution (e.g., Vonk et al. 2016; Goharrokhi et al. 2021). Intermediate-duration studies (e.g., Chmiel et al. 2020) capture seasonal and interannual variability but remain limited in resolving the high-frequency response of sediment

dynamics to extreme events. Collectively, these examples demonstrate how riverine inflows govern sediment fluxes, biogeochemical cycling, and contaminant transport in lakes and their surrounding ecosystems, while also highlighting important limitations in the temporal resolution and duration of existing observations.

Lake Geneva and the Rhône River provide an excellent natural laboratory to study these processes. The Rhône delivers approximately 1.7 million tons of SPM annually, ~90% of which originates from its catchment, while the remaining 10% comes from its other tributaries, including the Dranse, Venoge, Aubonne, and Versoix rivers (Loizeau and Dominik 2000; Loizeau et al. 2012). During stratified periods, the Rhône River forms a dense interflow within the thermocline (Piton et al. 2024), supplying nutrients that support surface productivity (Cotte et al. 2023). During unstratified periods, the river inflow plunges deeper into the lake, transporting nutrients and oxygen into the hypolimnion. Under extreme hydrological conditions, Rhône density currents can transport large particles, primarily within the silty–sandy fraction, through the sublacustrine canyon to distal accumulation zones (Silva et al. 2019). Meanwhile, finer particles within the interflow, including calcite and dolomite crystals typically ranging from 5 to 10 μm , can contribute to whiting events (Escoffier et al. 2022). Flocculation processes further modulate particle dynamics near the river mouth (Piton et al. 2024).

At the global scale, SPM variability reflects the interplay between fluvial sediment inputs, wind-driven resuspension, precipitation-driven runoff, and biological production, with the relative importance of each process depending on lake type and environmental setting (Cao et al. 2017; Cui et al. 2026). Recent global analyses further distinguish dominant regimes controlling SPM dynamics: wind-induced resuspension in shallow lakes, precipitation-driven inputs in deeper systems, and hydrologically controlled conditions in river-influenced lakes (Cui et al. 2026).

Within this framework, riverine inputs are generally the primary driver of SPM variability in large lakes with major inflows, particularly during high-discharge events when sediment-laden plumes enter the lake and control spatial and temporal gradients in suspended matter (Scheu et al. 2015; Zhu et al. 2024). However, internal lake processes, including wind-driven mixing, basin-scale circulation, and resuspension, can substantially redistribute particles and sustain elevated SPM levels beyond periods of direct river forcing (Valipour et al. 2017; Hofmann et al. 2011). Biological production and carbonate precipitation further contribute to the particle pool, particularly in productive and hardwater systems (Dittrich and Obst 2004; Escoffier et al. 2022).

Despite these advances, important limitations remain in our ability to resolve sediment dynamics across temporal scales. Long-term, high-frequency observations of SPM

dynamics in large lakes remain scarce, particularly datasets that capture the response of sediment fluxes to extreme hydroclimatic events. Most existing studies are based either on short-term campaigns or on temporally integrated measurements that smooth out event-scale variability. While remote sensing provides valuable insights into spatial patterns of surface SPM, it is generally limited to near-surface observations and lacks information on vertical fluxes. Together, these limitations restrict our ability to understand how floods, droughts, and storms alter sediment transport and deposition throughout the water column over time. This lack of multiyear, event-resolving observations represents a critical gap in predicting how climate-driven extremes will impact sedimentation and biogeochemical cycling in stratified lake systems.

To address this gap, the present study quantifies the vertical and temporal variability of export fluxes of organic matter, calcite precipitates, and minerals over a 5-year period based on regularly resolved sediment trap measurements (biweekly in summer and monthly in winter) in a large peri-alpine, river-influenced lake. These measurements are complemented by continuous high-frequency environmental observations from the LÉXPLORE platform, allowing us to evaluate the impact of extreme events on sediment flux dynamics. We hypothesize that extreme hydroclimatic events exert a strong control on particle fluxes, but that their effects vary depending on river forcing, inflow dynamics, and lake hydrodynamic conditions. We further expect that different types of extreme events produce contrasting sedimentation responses, reflecting variations in particle transport pathways and in-lake processes such as resuspension and calcite precipitation.

Study site

Lake Geneva and the LÉXPLORE platform

Lake Geneva, the largest freshwater lake in Western Europe, covers 580 km² with a maximum depth of 309 m and drains a catchment of about 8300 km². While glaciated alpine regions strongly influence its hydrology and sediment inputs, certain sub-catchments dominate sediment delivery. The Illgraben, one of the most active torrents in the European Alps, covers only ~0.1% of the Rhône basin yet contributes ~5–15% of its sediment load, exporting on average ~100,000 m³ year⁻¹ (~180,000–220,000 t year⁻¹) with large interannual variability, and typically generates two to seven debris flows or debris floods per year (Hirschberg et al. 2019; de Haas et al. 2020).

Our study was conducted at the LÉXPLORE platform, a floating research station anchored 570 m from the shoreline at a depth of 110 m, located at 46°30'0.819" N, 6°39'39.007"

E on Lake Geneva (Wüest et al. 2021). This infrastructure facilitates stable, year-round access for detailed sediment trap measurements and ensures the collection of a comprehensive, high-frequency dataset (Fig. 1).

Rhône River

The upper Rhône is a large river approximately 140 km long, originating at an elevation of ~2200 m. It serves as both the primary inflow and outflow of Lake Geneva. The Rhône delivers mineral-rich glacial meltwater and suspended sediments, which significantly influence the lake's hydrology, water quality, turbidity, and nutrient dynamics. Additionally, the Rhône contributes to thermal regulation, affecting seasonal stratification and circulation patterns within the lake. Continuous Rhône discharge and time-integrated suspended sediment samples (collected proportional to discharge over one week intervals) from Porte du Scex station, 5 km upstream of the river mouth, were provided by the NAWA-Fracht: national monitoring programme for substance load in Swiss watercourses (2025) (<https://doi.org/10.25678/000EKQ>).

Methods

Sediment traps

Sinking particulate samples were collected from January 2020 to December 2024 using cylinder sediment traps with an aspect ratio of 10.9 (Fig. S1), deployed near the LÉXPLORE platform at four depths (10, 30, 50, and 100 m). Traps were retrieved twice monthly from June to September and once monthly during the rest of the year. Trap contents were then freeze-dried and weighed. The mass flux is then calculated as:

$$\text{massflux} = \frac{\text{drymass}}{S \times \Delta d} \quad (1)$$

with the mass flux in g m⁻² day⁻¹, dry mass in g, $S = 0.0065$ m² (the area of the trap), and Δd the duration of exposition in days. Annual mean fluxes were adjusted to account for the varying exposure times. At certain times, gaps occurred between sampling periods, the most significant being from 11 November 2021 to 9 May 2022.

Samples processing and chemical analyses

After weighing, sediment samples were hand milled and analysed for total nitrogen (TN), total phosphorus (TP), organic carbon (TOC), inorganic carbon (TIC), and major elements (Ca, Mg, Al, Fe, K, Mn, Ti, Sr, Ba) at Eawag. Total



Fig. 1 Map of Lake Geneva and the Rhône catchment. The location of the Rhône inflow (white circle), the LéXPLORE platform (red circle) and the monitoring stations Porte du Scex (red circle along

the Rhone River). The inset highlights the distance from the Rhône inflow to the LéXPLORE platform, which is ~19 km

carbon (TC) and TN were measured using a combustion CNS elemental analyser (SoliTOC). TIC was determined following acidification with 3 M HCl, and TOC was calculated as the difference between TC and TIC. For TP analysis, sediment samples were digested for 2 h at 121 °C with potassium persulfate and measured photometrically using a flow-injection analyser (FIA, Procon AG). The major elements (Mg, Al, K, Na, Ca, Fe, Ti, Cr, Mn) were quantified by ICP–MS (Agilent 7900) after microwave-assisted aqua regia digestion of sediment samples, followed by dilution, centrifugation, and calibrated against certified standards. The flux for each element was calculated as the concentration times the total mass fluxes.

Mass balance for the different fractions

From the individual fluxes, we estimated contributions from biomass, calcite precipitation, and mineral oxides. Biomass flux was calculated as the sum of TOC, TN, TP, and oxygen and hydrogen, contributions inferred from Redfield

stoichiometry ($O:C = 110:106$ and $H:C = 263:106$) to approximate total organic matter mass and facilitate comparison with bulk biomass fluxes. Calcite flux was derived from TIC expressed as $CaCO_3$. Masses of individual minerals were converted into corresponding mineral oxides typical of the Rhône sediments (Stutenbecker et al. 2018), including MgO , Al_2O_3 , K_2O , TiO_2 , Cr_2O_3 , MnO , and Fe_2O_3 .

LéXPLORE dataset

The LéXPLORE platform provides an extensive high-frequency dataset of background environmental data, openly accessible on <https://www.datalakes-eawag.ch>. In this study, we used data from vertical profiles obtained with multiparameter sondes (Sea and Sun Technology), recording temperature, oxygen, turbidity, and conductivity (<https://www.datalakes-eawag.ch/datadetail/674>).

Thermal structure was further characterized using data from the thermistor chain over a 5-year period (<https://www.datalakes-eawag.ch/datadetail/448>). To investigate extreme

events, we additionally analyzed current velocities (<https://www.datalakes-eawag.ch/datadetail/599> and <https://www.datalakes-eawag.ch/datadetail/600>), wave height (<https://www.datalakes-eawag.ch/datadetail/885>) and the meteorological data from the platform station (<https://www.datalakes-eawag.ch/datadetail/459>).

Rhône River intrusion depths and total suspended solids

The Rhône is monitored at high frequency at Porte-du-Scex, approximately 5 km upstream from its mouth. Real-time hydrological data are obtained through the Linked Data Services platform (<https://www.lindas.admin.ch>). Water samples are collected continuously throughout each week and are proportional to flow. Total suspended solid (TSS) input from the Rhône are calculated by multiplying the TSS concentrations measured in the weekly composite samples by the river discharge over this time period, thereby integrating TSS over the weekly period.

Continuous measurements of water temperature and conductivity were used to estimate the nominal intrusion depth, following the method described by Cortés et al. (2014a). The intrusion depth represents the level at which the inflowing river water reaches density equilibrium with the stratified lake water. The intrusion depth is calculated by comparing the density profiles of the Rhône River and the lake to find the depth where they match within a specified tolerance. The tolerance values range from $5\text{E}-5 \text{ kg m}^{-3}$ up to $5\text{E}-4 \text{ kg m}^{-3}$ depending on the considered year and data availability.

For the Rhône River, salinity is estimated from in situ conductivity and temperature measurements. Conductivity (C) values (in $\mu\text{S cm}^{-1}$) are first normalized to a reference temperature of 20°C using a temperature correction function, then converted to salinity using a calibrated proportionality constant ($y_{\text{cond}} = 0.000874$). The salinity S (in g kg^{-1}) is thus calculated as:

$$S = y_{\text{cond}} \cdot f(T) \cdot C$$

$$\text{w h e r e } f(T) = 1.8626 - 0.052908 \cdot T + 0.00093057 \cdot T^2 - 6.78 \cdot 10^{-6} \cdot T^3.$$

For the lake, salinity values were already available through the LÉXPLORE data and had been obtained using the same conductivity–temperature–based calculation method.

Water densities are calculated as a function of temperature and salinity, using the formula developed by Chen and Millero (1986):

$$\begin{aligned} \rho(\text{g/cm}^3) = & 0.9998395 + 6.7914 \cdot 10^{-5}T - 9.0894 \cdot 10^{-6}T^2 \\ & + 1.0171 \cdot 10^{-7}T^3 - 1.2846 \cdot 10^{-9}T^4 \\ & + 1.1592 \cdot 10^{-11}T^5 - 5.0125 \cdot 10^{-14}T^6 \\ & + (8.181 \cdot 10^{-4} - 3.85 \cdot 10^{-6}T + 4.96 \cdot 10^{-8}T^2) \cdot S \end{aligned}$$

Hydrodynamic simulation with Alplakes

The Alplakes website (<https://www.alplakes.eawag.ch/>) employs numerical models to simulate physical processes in lakes, with a particular focus on vertical mixing and thermal stratification. In this study, the simplified particle-tracking module of Alplakes is used to simulate particle behaviour over time. The model allows particles to be introduced at defined times, locations, and intrusion depths, with control over their number and spatial distribution, and simulates their transport pathways within the lake based on meteorological forcings.

While more advanced offline particle-tracking modules exist that allow detailed trajectory analysis and visualizations as well as fully quantitative tracking, the online version adequately captures the dispersal patterns of Rhône River particles. This functionality is particularly useful for examining how particles originating from the Rhône River might disperse upon entering the lake.

Results

Lake stratification and meteorological conditions

The seasonal variation of water temperature with depth from 2020 to 2024 shows clear annual patterns of thermal stratification. Stratification typically develops from April to October, with warm surface waters and a distinct thermocline (Fig. 2). In winter, the water column becomes colder and more uniform as stratification breaks down. A comparison across years highlights 2022 as the warmest year, with surface temperatures highest and the thermocline extending deepest, reaching about 25–30 m. In contrast, 2020, 2023, and 2024 display similar conditions but with slightly shallower thermoclines and cooler surface waters. Notably, 2021 recorded the coolest temperatures overall, and stratification that year was not continuous but was disrupted twice. Overall, the data reveal pronounced seasonal cycles and moderate interannual differences.

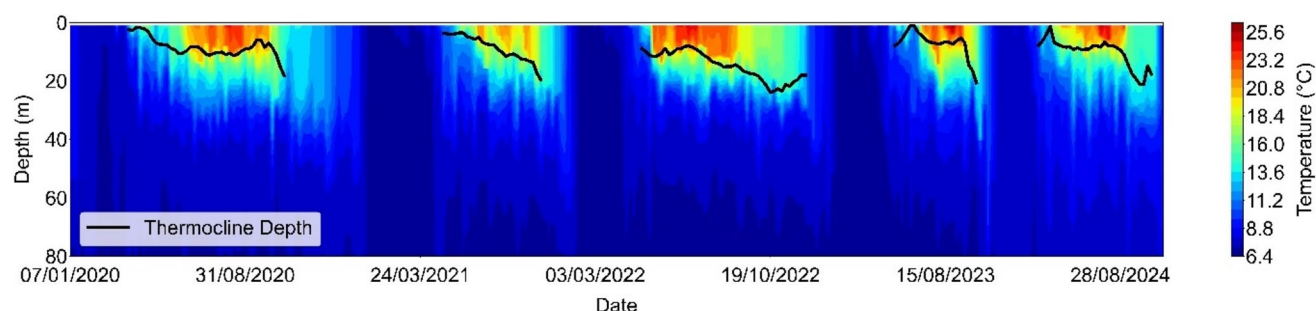


Fig. 2 Water temperature from 2020 to 2024 at LéXPLORE Platform. Black lines represent the thermocline depth. Data are from DATALAKES (<https://www.datalakes-eawag.ch/datadetail/802>)

Table 1 Total annual input of water and total suspended solids brought by the Rhône River

Year	Total Rhône discharge (km ³ year ⁻¹)	Total suspended solids (kt year ⁻¹)	TSS from June to August (% of annual)
2020	6.21	2159	54
2021	6.10	2175	89
2022	5.23	1751	74
2023	6.39	1973	69
2024	8.04	6059	79
mean	6.39	2823	73

Annual and seasonal inputs from the Rhône River

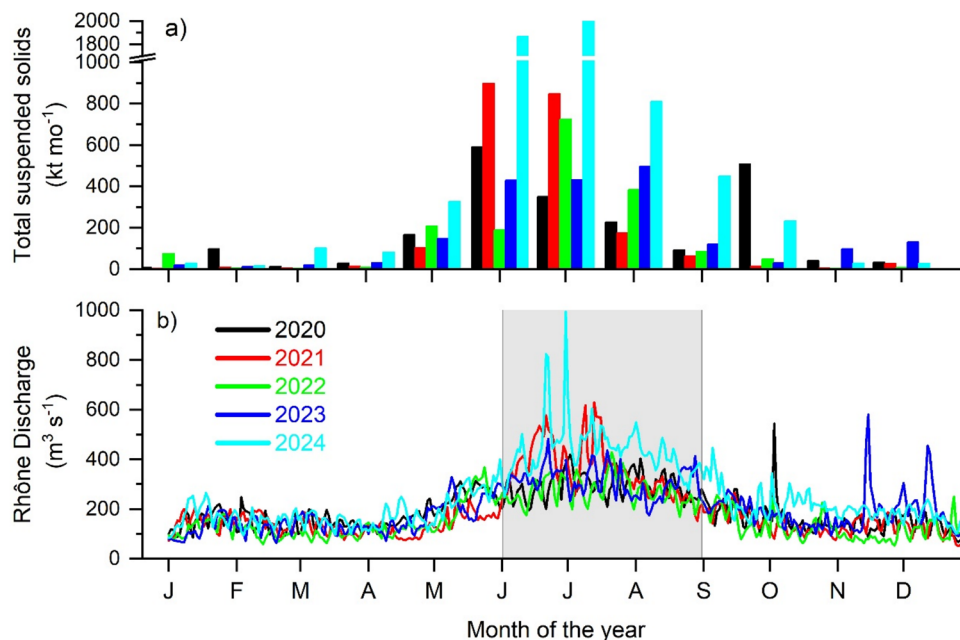
The total annual discharge of the Rhône was similar for the years 2020, 2021, and 2023 (Table 1). It was lower in 2022 and significantly larger in 2024, reaching a value

about 34% greater than the mean of the previous years. The TSS amounts for 2020–2023 remained similar, with the lowest value recorded in 2022. However, the value in 2024 increased dramatically, reaching approximately three times the average of the previous four years.

Data are compiled from the monitoring station Porte du Scex (NAWA-Fracht: <https://doi.org/10.25678/000EKQ>).

Seasonal fluctuations of the Rhône discharge (Fig. 3b) are generally similar, with the highest flows occurring from June to August. However, the years 2021 and 2024 recorded higher flows than usual summers, with two extreme discharge events at the end of June and the beginning of July 2024. In 2023, the winter was exceptionally wet, with significant discharges in November and December. TSS (Fig. 3a) follow a similar pattern to the discharge, with maximum inputs in June, July, and August, and a total of about 73% of the annual TSS (Table 1). Significant inputs also occurred in

Fig. 3 Annual Rhône discharge and total suspended solids. Inputs of TSS are shown in kilotons per month at Porte du Scex (panel a). The break in the y axis extends from 1010 until 1700 kt month⁻¹. Panel b shows Rhône discharge during the 5 years. The peak in discharge from 1 June to 31 August is highlighted in grey



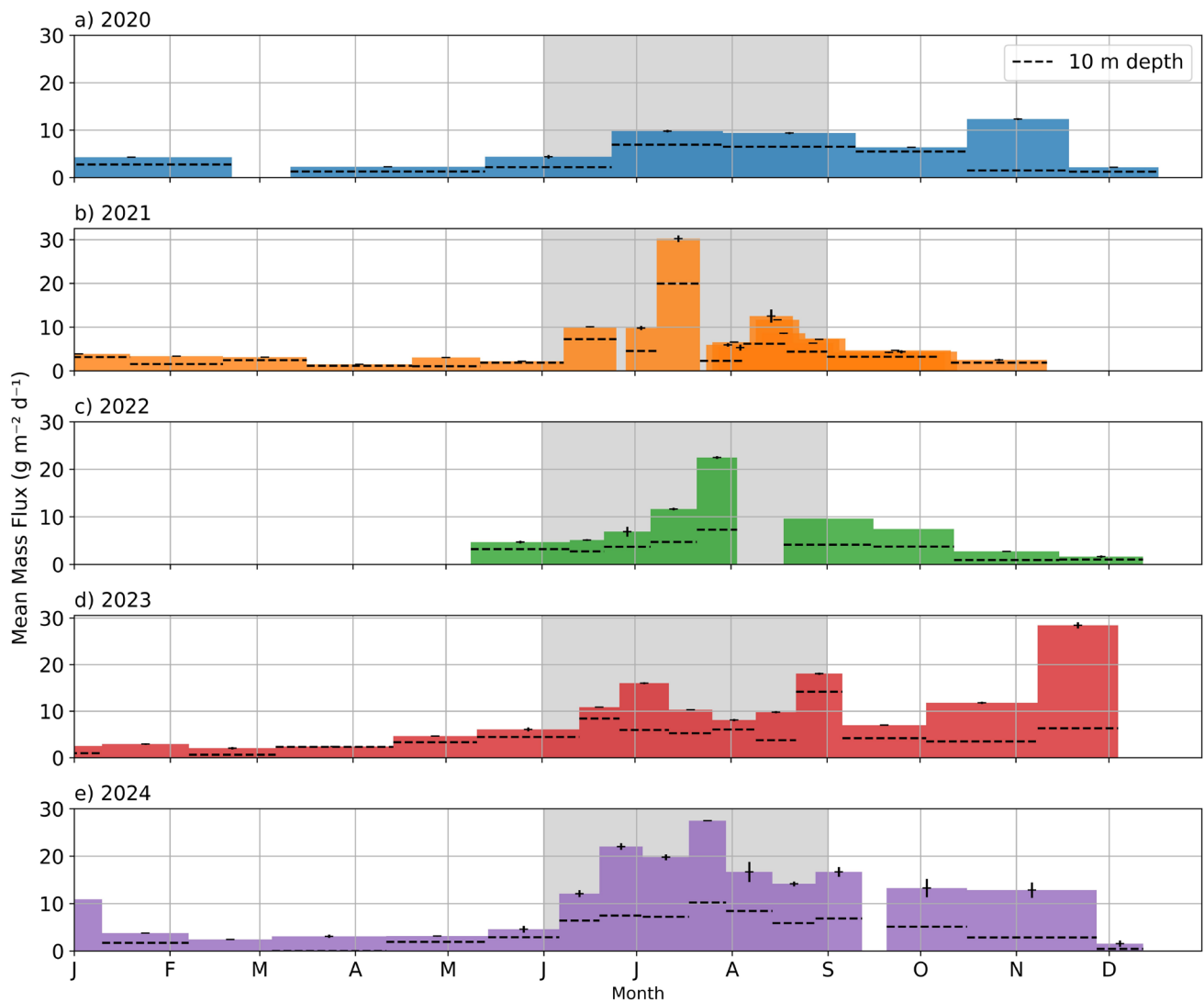


Fig. 4 Total mass fluxes ($\text{g m}^{-2} \text{day}^{-1}$) over the 5 years. Averaged values are presented for 30, 50, and 100 m depths (colored bars) in 2020 (a), 2021 (b), 2022 (c), 2023 (d), and 2024 (e). The black

dashed line represents the total mass flux at 10 m depth. The peak in discharge from 1 June to 31 August is highlighted in grey. Note the data gaps, for example from 11 November 2021 to 9 May 2022

October 2020, in November and December 2023 (Fig. 3a). The lowest values overall were recorded in 2022.

Seasonal and vertical variations in sinking particulate fluxes

Sinking particulate fluxes in Lake Geneva showed marked interannual variability over the 5-year period (Fig. 4). Total mass fluxes were highest in 2023 and 2024 ($2.63 \text{ kg m}^{-2} \text{ year}^{-1}$ each) and lower in 2020 ($1.80 \text{ kg m}^{-2} \text{ year}^{-1}$), 2021 ($2.07 \text{ kg m}^{-2} \text{ year}^{-1}$), and 2022 ($2.12 \text{ kg m}^{-2} \text{ year}^{-1}$). Throughout the 5 years, annual total mass fluxes generally peaked between June and September (Fig. 4). While fluxes from January to May were consistently lower, autumn values were variable: they were relatively low

in 2021 and 2022 but exceeded summer peaks in 2020 and 2023. During the inundations of June 2021, particulate fluxes reached $30 \text{ g m}^{-2} \text{ day}^{-1}$. Elevated fluxes also occurred during the June–July 2024 floods and in November 2023. The observed fluxes at 30, 50, and 100 m were homogeneous (Fig. S2). Therefore, we used the average of the three fluxes in all subsequent analyses presented in this manuscript. At 10 m, fluxes were systematically lower than the depth-average at 30, 50, and 100 m, particularly during high-flux events (Fig. 4).

Consistent with total mass, fluxes of TN, TP, TOC, and TIC were generally higher below 30 m than at 10 m, with the largest differences during high-flux events (Fig. S3). Exceptions occurred in July–August of most years, when elevated fluxes were observed in the upper water column.

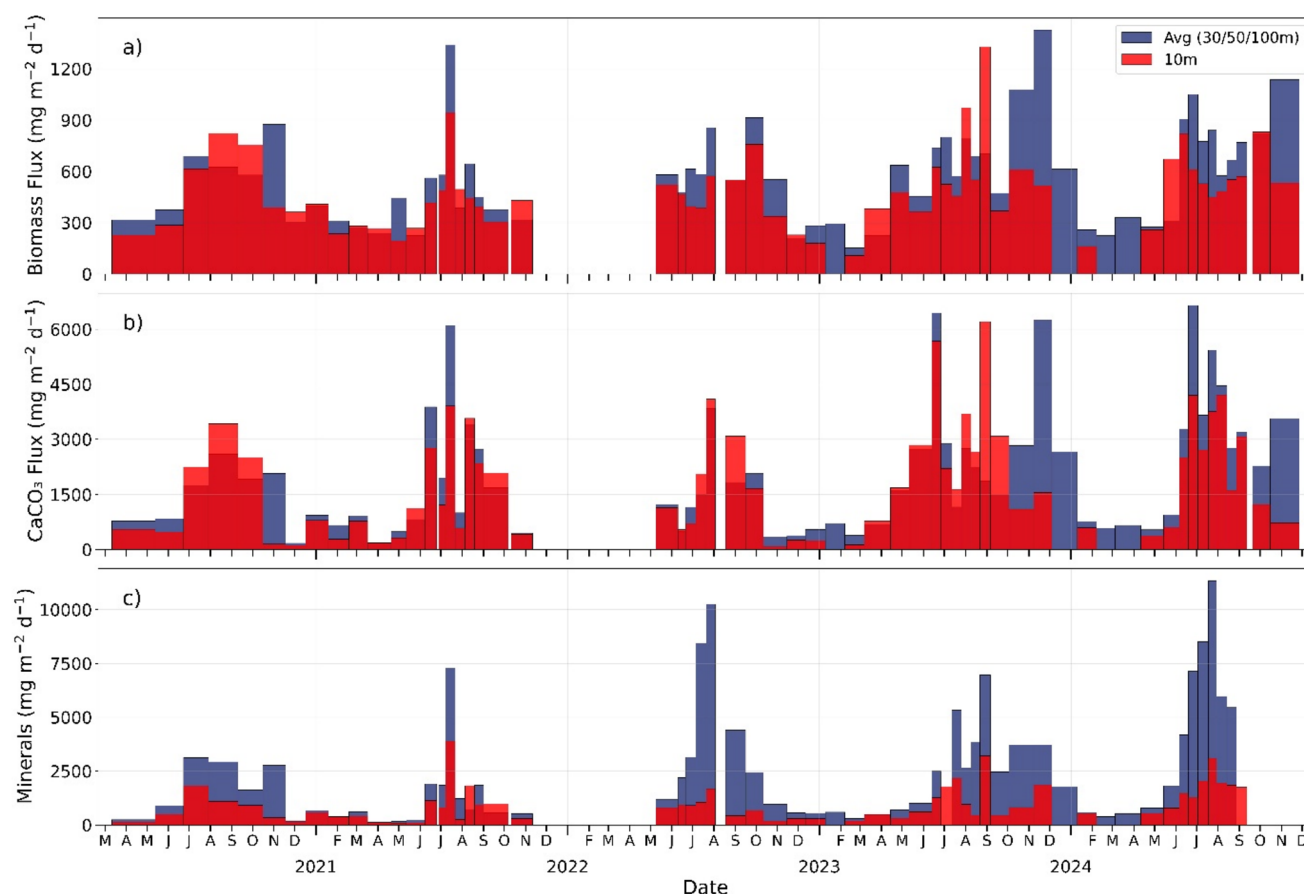


Fig. 5 Phase-specific fluxes (in $\text{mg m}^{-2} \text{ day}^{-1}$) across the 5 years. Fluxes at 10 m (red) and average fluxes from 30, 50, and 100 m (blue) are shown for the biomass (a), calcite (b), while mineral fluxes (c) correspond to measurements at 30 m. Bar widths reflect the duration of sediment trap deployments (time between deployment and recov-

ery). Gaps in the 10 m record reflect periods when the shallow sediment trap was lost during handling, while deeper traps continued to operate. Note the absence of data from 11 November 2021 to 9 May 2022, as well as smaller gaps associated with variations in sampling

Seasonal peaks coincided with thermal stratification, and elevated values frequently extended into autumn. Biomass fluxes showed the same seasonal pattern and similar magnitudes across depths ($0.17\text{--}0.23 \text{ kg m}^{-2} \text{ year}^{-1}$) (Fig. 5a).

Calcite fluxes were substantial, with elevated values during summer stratification and late-season productivity, including a peak of $6 \text{ g m}^{-2} \text{ day}^{-1}$ in November 2023. Across years, mean calcite fluxes at 10 m ($0.71 \pm 0.53 \text{ kg m}^{-2} \text{ year}^{-1}$) and at 30–100 m ($0.67 \pm 0.65 \text{ kg m}^{-2} \text{ year}^{-1}$) were comparable within variability, and in 2022, 2023, and 2024 the 10 m traps had higher means than the deeper traps (Fig. 5b).

Mineral fluxes showed consistently higher values below 30 m and strong increases during peak runoff events. The mean annual flux at 10 m was $0.37 \pm 0.32 \text{ kg m}^{-2} \text{ year}^{-1}$ and $0.98 \pm 1.01 \text{ kg m}^{-2} \text{ year}^{-1}$ below 30 m.

Intrusion depth of the Rhône River interflow

Throughout the year, the intrusion depth of the Rhône River interflow varied markedly, ranging from approximately 10 m in May to about 40 m (Fig. 6). During spring, intrusions were generally shallower, reaching depths of 10–20 m, while in summer they typically extended to 20–30 m. The vertical turbidity profiles acquired at LÉXPLORE clearly reveal the Rhône River interflow during the stratified period.

Hydrodynamics of extreme events

During summer 2021, two major flooding periods were identified (Fig. 7a–c): 17–27 June and 13–27 July. Corresponding turbidity peaks were observed near the LÉXPLORE site during both events. On 24 June, turbidity increased markedly at approximately 9 m depth following the sustained rise in Rhône discharge from 15 to 24 June. Similarly, on 16 July,

Fig. 6 Tracking the intrusion depth of the Rhône River inter-flow. Calculated weekly Rhône River intrusion depth in Lake Geneva from 2020 to 2024 is shown as a white line. Turbidity profiles from the multiparameter sonde at LÉXPLORE are averaged over time and depth for the same period

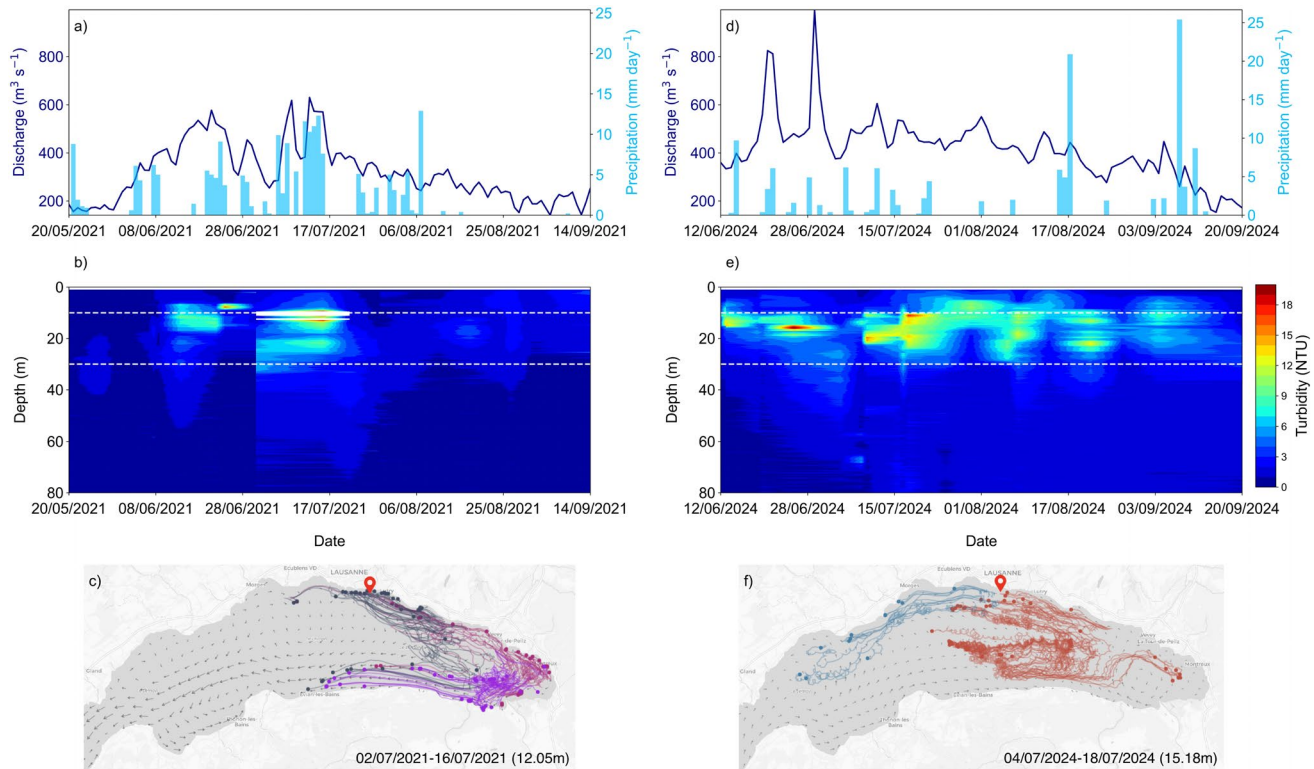
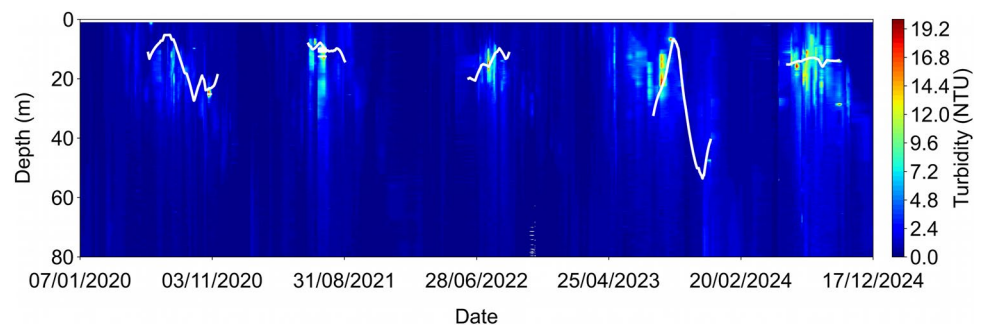


Fig. 7 Tracking flood event inputs in 2021 and 2024. Rhône discharge and rainfall (a, d), turbidity (b, e) and Alplake simulation (c, f) are shown during summer 2021 (left panels), summer 2024 (right panels). The LÉXPLORE platform is shown as the red icon in panels c and f. In panel c, particles released on 9 July are shown in pink, while those released on 13 July are shown in purple. Additional particles (grey) were released on 13 July to represent a subset of the particles initially released on 9 July that are transported toward the LÉX-

PLORE platform, allowing clearer visualisation of their trajectories. In panel f, two simulations were performed: an initial simulation from 21 June to 4 July to represent the precipitation peak, followed by a second simulation from 4 to 18 July (shown here) to capture the turbidity peak at the LÉXPLORE platform. Blue particles released on 4 July were initialised from the positions of selected particles originally introduced near the Rhône on 21 June, while particles released on 30 June are shown in red

turbidity peaked around 12 m depth, coinciding with major discharge events on 9, 13, and 16 July.

To track particle dispersion within the lake between 9 and 16 July, simulations were performed using the Alplakes hydrodynamic model at the corresponding depth. Particles released on 9 and 13 July followed distinct pathways, with a subset of 9 July particles extended to illustrate their

continued transport toward the LÉXPLORE platform. While many particles remained within the eastern basin, those transported further into the lake followed the northern shore, ultimately reaching the LÉXPLORE site.

During the floods in summer 2024, the turbidity data show distinct peaks from mid-June to the end of August that align with increased Rhône inflows (Fig. d, e, f). Between

mid-June and mid-July, the calculated intrusion depth was approximately 15 m, matching the depth of peak turbidity. Due to the two-week limitation of the Alplakes model, the simulations were performed in two consecutive periods. The first simulation, from 21 June to 4 July, captured the high-discharge peak of the Rhône, while the second, from 4 to 18 July, tracked the subsequent trajectories of these particles. The results show that particle inputs on 21 June and 30 June contributed to a substantial arrival of particles at the LÉXPLORE site by mid-July. Overall, the Rhône-derived particle load was distributed between the northern and southern shores of the lake.

During the 2022 summer drought, turbidity data show a peak on 25 July, centred at approximately 15 m depth, while the Rhône intrusion depth at that time was estimated at ~20.4 m (Fig. 8a, b). The highest Rhône discharge occurred between mid-July and the end of July, with a marked peak on 21 July. According to the Alplakes model, particle transport in 2022 was strongly shaped by two circulation features: a small gyre near the Rhône delta, which temporarily trapped particles, and a larger gyre in the central

basin (Fig. 8c). In this configuration, the Rhône intrusion rotated within the small clockwise gyres and the large anti-clockwise gyre, with a fraction of particles escaping from the delta region on 23 July and reaching the LÉXPLORE site.

From October to December 2023, a sequence of intense storms controlled particle dynamics in Lake Geneva. Strong fluctuations in Rhône discharge reflected multiple heavy rainfall events during this period (Fig. 8d). Wind, waves, and current measurements at LÉXPLORE showed pronounced activity throughout the autumn–winter season (Fig. S5). Distinct turbidity pulses occurred at mid-depths in October and November, and a major peak was recorded on 13 December at approximately 48 m depth (Fig. 8e). Modeled particle trajectories indicate that circulation in the eastern basin was dominated by a basin-scale gyre, with particles following both recirculating pathways and transport along the northern shore (Fig. 8f).

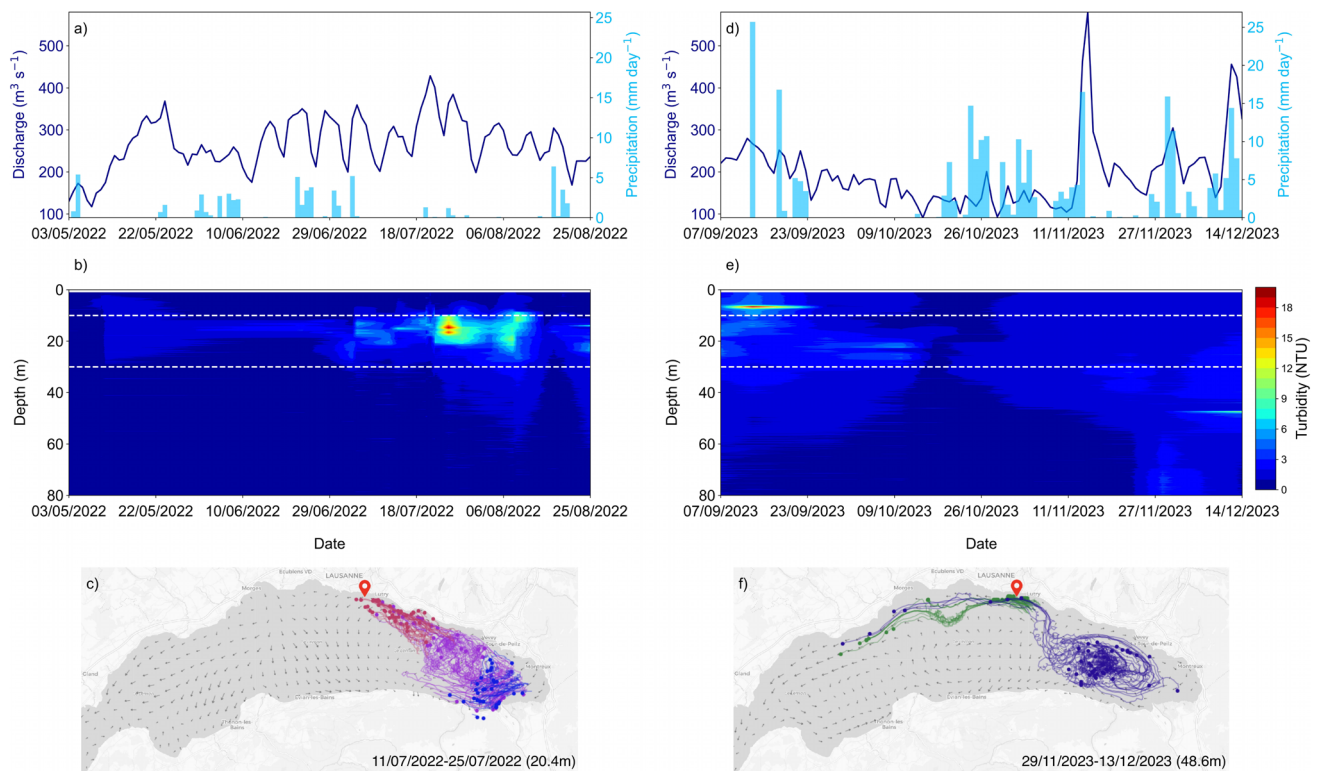


Fig. 8 Impact of additional extreme events on particles within Lake Geneva. Rhône discharge and rainfall (a, d), turbidity (b, e) and Alplake simulation (c, f) during drought conditions in summer 2022 (left panels) and extreme storms during autumn–winter 2023 (right panels). In panel c, particles released on 11 July (purple) represent transport over the full simulation period. The Rhône discharge peak on 21 July is represented by particles shown in blue, while particles

released on 23 July (pink) correspond to a subset of those from 21 July, selected to illustrate the trajectories of particles transported toward the LÉXPLORE platform. In panel f, particles shown in green correspond to positions at the end of the initial simulation (15–29 November), while particles shown in blue were introduced on 29 November and tracked together with them until 13 December

Discussion

Drivers of sinking particulate flux dynamics

Sediment dynamics in lakes are controlled by the interplay between external forcing and internal processes, including fluvial inputs, wind-driven resuspension, and biological production. The relative importance of these drivers varies across lake systems depending on hydrology, morphology, and catchment characteristics.

In large lakes with major inflows, riverine inputs are typically the dominant driver of suspended matter variability, particularly during high-discharge events when sediment-laden plumes enter the lake and control spatial and temporal gradients. This dominant role has been documented across large lake systems, including Lake Superior, where suspended matter inputs strongly regulate nearshore biogeochemistry and track river discharge (Hudson et al. 2024). However, internal lake processes such as wind-driven mixing, hydrodynamic circulation, and resuspension can substantially redistribute particles and maintain elevated suspended matter beyond periods of direct fluvial input (Valipour et al. 2017; Hofmann et al. 2011). In addition, biological production and carbonate precipitation further modulate particle composition and fluxes during productive periods (Dittrich and Obst 2004; Escoffier et al. 2022).

Within this framework, Lake Geneva represents a large, river-influenced lake where sediment-trap observations demonstrate that sinking particulate fluxes are shaped by fluvial inputs, internal primary production, and carbonate precipitation, further modulated by resuspension and hydrodynamic transport. Strong riverine forcing interacts with seasonal stratification and internal dynamics to regulate both horizontal redistribution and vertical particle export. These processes control the timing, magnitude, and vertical distribution of particle export throughout the water column. By resolving vertical fluxes and particle composition, our observations provide a process-oriented perspective on sediment dynamics that complements surface-based approaches.

Our results provide a multiyear perspective on how contrasting extreme events modify sedimentation processes through changes in forcing, transport pathways, and particle supply. Fluvial inputs from the Rhône River strongly influence particle fluxes and mineral delivery. In the present record, periods of elevated Rhône River discharge coincide with increased total mass and mineral fluxes at depth, with a clear step between 10 and 30 m that aligns with the Rhône intrusion layer near 10–40 m (Figs. 4, 5, 6). This vertical structure highlights the role of density-driven intrusions in controlling not only horizontal transport but also the depth of particle injection into the water column. Shallow

sediment-trap measurements therefore extend previous observations from earlier deeper deployments (≥ 60 –75 m depth), which also documented strong seasonal patterns linked to Rhône River inputs (Gandais and Vernet 1987; Gandais 1987; Dominik et al. 1993; Graham et al. 2016; Gascon Diez et al. 2017). Similarly, accumulation rates decrease away from the Rhône River mouth toward the basin centre (Loizeau et al. 2012), a spatial pattern observed in other lake systems.

Beyond fluvial inputs, the composition of sinking fluxes also reflects internal lake processes, particularly carbonate precipitation. Calcite contributes approximately 40% of total flux and peaks during stratified, productive periods, in agreement with prior high-frequency observations at the LÉXPLORE platform (Escoffier et al. 2023). Calcite fluxes were often higher at 10 m in July and August (Fig. 5b), consistent with enhanced carbonate saturation associated with higher pH from primary production and warmer temperatures (Dittrich and Obst 2004). Episodic whiting events, massive calcite precipitation episodes, have been recurrently reported in Lake Geneva (Escoffier et al. 2022; Many et al. 2022), typically associated with high Rhône discharge, warm surface waters (> 15 °C), and strong stratification. The unusually high calcite fluxes observed in November 2023 likely reflect resuspension of carbonate-rich nearshore sediments during strong storms and wave action rather than in situ precipitation (see 6.3).

Organic (biomass) fluxes varied little with depth (Fig. 5a), but this masks ongoing consumption and transformation within the water column. Oxygen consumption, a proxy for organic matter mineralization, indicates that organic matter produced in the surface layers is actively remineralized throughout the water column (Schwefel et al. 2018; Müller et al. 2019; Steinsberger et al. 2021). This consumption may be compensated by sediment focusing, which enhances deeper fluxes through lateral redistribution (Sturm and Matter 1978; Finger et al. 2006). Historical trap studies recorded seasonal peaks in organic carbon fluxes during productive periods (Gandais and Vernet 1987), while recent LÉXPLORE observations confirm primary production as the dominant driver of organic carbon export (Fernández Castro et al. 2021a; Steinsberger et al. 2021; Carratalà et al. 2023; Escoffier et al. 2023). Together, these processes indicate that organic fluxes reflect the balance between production, transformation, and physical redistribution within the water column.

Similar process interactions have been reported in other large lake systems. Comparable spatial and temporal sedimentation patterns have been found in Lakes Onego, Michigan, Winnipeg, Brienz, Maggiore, and Constance, where riverine inflows play a dominant role in shaping lacustrine sediment distributions and biogeochemical cycling (Fuentes et al. 2013; Scheu et al. 2015; Fink et al. 2016; Fabbri et al.

2021; Goharrokhi et al. 2021; Belkina et al. 2023; Zhu et al. 2024). Together, these cross-system comparisons confirm that the coupling between river discharge, particle transport, and basin-scale sedimentation observed in Lake Geneva reflects a general feature of large, river-influenced lakes.

Overall, particulate flux dynamics in Lake Geneva emerge from the combined influence of riverine inputs, biological production, and carbonate precipitation, all modulated by hydrodynamic circulation that couples nearshore and pelagic zones. More broadly, this combination of external forcing and internal processes is characteristic of many large, stratified lakes, where the balance between fluvial inputs, in-lake production, and physical redistribution governs both sediment transport and biogeochemical cycling. This integrative view emphasizes that vertical particle fluxes cannot be interpreted from surface conditions alone but require accounting for subsurface transport and transformation processes.

Sediment dynamics within the river interflows

Sediment dynamics within river intrusions in stratified lakes are primarily controlled by density gradients, which determine the depth and persistence of particle transport pathways. When a sediment-laden inflow enters a stratified water column, its relative density, governed by temperature, dissolved substances, and suspended sediment concentration, controls whether it forms an overflow, interflow, or underflow, thereby regulating how and where particles are transported and deposited. Under stratified conditions, the pycnocline traps intrusions at their level of neutral buoyancy, promoting lateral transport and mid-depth sedimentation while limiting vertical exchange. Mixing at the river mouth, including plunging and entrainment, rapidly modifies inflow density and thus its trajectory and transport capacity.

During the stratified period in Lake Geneva, the strong density gradient at the thermocline acts as an effective barrier controlling the depth at which the Rhône River interflow intrudes (Blanckaert et al. 2024). The interflow typically follows the seasonal deepening of the thermocline through summer and autumn (Fig. 6). Intense shear and turbulence near the river mouth promote rapid flocculation within the interflow's mixing layer up to 1 km from the outlet (Piton et al. 2024). Recent concurrent sediment trap data from the Rhône River delta and deep basin in Lake Geneva (Mittelbach et al. 2025) complement our observations for 2022, confirming spatial differences in organic and inorganic carbon fluxes related to river influence and lake productivity. At the LÉXPLORE site, interflow dynamics are modulated by meteorological forcing and large-scale circulation (Imboden and Wüest 1995; Cortés et al. 2014b). During storm events, increased discharge generates underflow currents that propagate along the lake bottom, transporting sediments and solutes

to deeper layers (Wells and Dorrell 2021). Under typical conditions, however, river water intrudes at the thermocline, spreading laterally where its density matches that of ambient lake water and transporting sediments along this interface (Soulignac et al. 2021; Piton et al. 2022). Based on modelling, such interflows occur approximately 63% of the time, while denser underflows develop around 32% of the time and buoyant overflows only about 5% of the time (Lu et al. 2022). More recent measurements indicate an even stronger dominance of interflow behaviour, with the Rhône inflow being trapped within the pycnocline more than 98% of the time during the 23 September 2016 to 21 January 2017 observation period (Blanckaert et al. 2024). Occasionally, stronger inflows still penetrate below the thermocline, redistributing sediments and nutrients within the hypolimnion.

At the LÉXPLORE platform, 19 km downstream of the Rhône delta, elevated mineral fluxes below 30 m depth indicate that aggregation processes persist along the interflow pathway. Although direct floc measurements were not available, the combination of high sinking fluxes and similar magnitudes between 30 and 100 m suggests removal of fine particles through aggregation and rapid settling. The limited vertical variation in fluxes implies enhanced settling within a narrow density layer rather than simple depth stratification. Previous studies reported a logarithmic decrease in sedimentation rates with distance from the Rhône delta (Loizeau et al. 2012), reflecting progressively lower deposition offshore. Our observations align with this trend, suggesting that a large fraction of suspended minerals settles before reaching the central basin.

Similar density-driven interflows have been documented in other large perialpine and temperate lakes. In Lake Constance, turbidity currents and interflows from the Alpine Rhine transport fine sediments across the thermocline before deposition in distal basins (Sturm and Matter 1978; Fink et al. 2016). In Lake Lucerne, tributary density currents promote sediment focusing along basin slopes, with seasonal variability linked to runoff (Finger et al. 2006). Recent observations in Lake Geneva further highlight the ubiquity of such processes, where sediment-laden inflows from the Rhône generate recurring interflows that shape mid-depth sedimentation and water-column stratification (Lu et al. 2022; Blanckaert et al. 2024). Comparable density-driven intrusions and interflows have also been reported in many other lakes worldwide (Thorez et al. 2024). This widespread occurrence confirms that interflow-mediated transport is a fundamental mechanism in stratified lakes receiving significant fluvial inputs. Collectively, these analogs demonstrate that interflow-mediated transport and mid-depth sedimentation are common and recurrent features of stratified lakes receiving high sediment inputs from alpine or fluvial sources.

Together with our observations, these cross-system comparisons demonstrate that interflow-mediated transport and associated mid-depth sedimentation are not site-specific features, but rather represent a general mechanism in stratified lakes receiving significant fluvial inputs. This highlights the broader relevance of our findings for understanding sediment dynamics and the redistribution of particles, nutrients, and contaminants in similar lake systems.

Response of the Rhône River–Lake Geneva system to extreme events

Extreme hydroclimatic events influence sediment dynamics through distinct mechanisms, depending on hydrological forcing, stratification, and antecedent catchment conditions. The Rhône–Lake Geneva system demonstrates a pronounced sensitivity to extreme hydrometeorological conditions, which can substantially modify river discharge regimes, sediment transport pathways, and interflow dynamics within the lake basin. Such event-driven variability is increasingly recognized as a major control on annual sediment and nutrient budgets in large lakes. During the study period, a sequence of contrasting events underscored this sensitivity: severe floods in summer 2021 and again in 2024, along with violent thunderstorms occurring after an extended dry spell in July 2022 and during an already wet period in late autumn 2023. Together, these events provide valuable insight into how variations in precipitation, antecedent precipitation and hydrological conditions, and catchment hydrology govern turbidity, particle fluxes, and sediment deposition patterns in Lake Geneva. Similar responses to extreme hydrometeorological forcing have been reported in other peri-alpine lakes, highlighting the general importance of event-driven variability for sediment transport and deposition (Fink et al. 2016; Perga et al. 2018; Råman Vinnå et al. 2018).

The floods of 2021 and 2024 illustrate the combined effects of persistent heavy rainfall, high lake levels, and enhanced runoff on sediment dynamics. In 2021, prolonged rainfall across Switzerland caused widespread inundation, with high Rhône discharge and elevated lake levels driving extensive shoreline flooding and increased suspended sediment input. Similarly, in late June and July 2024, intense rainfall combined with rapid snowmelt and glacier melt generated peak Rhône flows of $\sim 1000 \text{ m}^3 \text{ s}^{-1}$, typical of a 100-year flood event. Interestingly, sediment fluxes measured at the LÉXPLORE were lower in 2024 than during the 2021 flood. This difference suggests that a significant fraction of sediment-laden water bypassed surface and intermediate layers as density-driven underflows. This interpretation is supported by unusual turbidity peaks recorded at 60 m depth (Fig S4), consistent with the formation of density-driven underflows previously documented in Lake Geneva and other deep lakes (Sturm and Matter 1978; Giovanoli 1990;

Corella et al. 2014; Thorez et al. 2024). Moreover, the lake's hydrodynamic conditions, including current direction, thermal stratification, and wind-driven circulation, influenced the transport pathways and transit time of the Rhône plume toward LÉXPLORE. These contrasting responses highlight that the structure of river intrusions, controlled by hydrological forcing and stratification, governs whether sediment is transported at mid-depth or transferred more directly to deeper layers.

In contrast to these large-scale floods, short-duration convective storms in July 2022 and November–December 2023 show how antecedent soil moisture critically shapes runoff and sediment mobilisation. The summer of 2022 was characterised by an exceptional drought, with extremely low Rhône discharge limiting the delivery of TSS. However, a series of intense thunderstorms produced localised but vigorous runoff over dry soils, leading to unexpected sediment pulses captured by sediment traps (Fig. S5). Concurrent strong wind events likely enhanced sediment resuspension in nearshore zones, as observed in other large lake systems such as Lakes Constance, Balaton and Turkana (Filstrup and Lind 2010; Hofmann et al. 2011; Zăinescu et al. 2023). By contrast, the late-autumn storms of 2023 struck an already saturated catchment, amplifying runoff generation and sediment supply to the nearshore lake zone. Six significant rain events occurred in late autumn 2023, accompanied by wind speeds averaging over 10 m s^{-1} on twelve occasions and a record wave height peaking at 1.12 m on 5 November, alongside five other events exceeding 0.8 m (Fig. S6). These energetic conditions likely mobilised large amounts of fine sediments from the shoreline and shallow zones. Similar processes have been described for Lake Erie, where wind-driven resuspension produces pronounced pulses of biogenic particles (Valipour et al. 2017). The resulting sedimentation rates were the second highest observed during the study period, exceeded only by those of the 2021 floods.

Sediment-trap data from autumn 2023 show pronounced peaks in biomass and calcite fluxes, whereas mineral fluxes display only a moderate increase (Fig. 5). At first glance, this pattern could suggest enhanced biological production. However, elevated chlorophyll-*a* concentrations were observed from 26 September to 11 October at the LÉXPLORE platform. After this period, phytoplankton biomass declined markedly, with substantially lower chlorophyll-*a* concentrations in November and December. This temporal decoupling clearly indicates that the elevated biogenic fluxes observed in autumn 2023 cannot be attributed to new production. Instead, this pattern suggests that storm-driven resuspension preferentially mobilized fine organic and carbonate particles rather than the heavier mineral grains typical of the sediment bed (Corella et al. 2014).

The exceptionally high biomass fluxes are therefore best explained by the re-entrainment of legacy biogenic material, combined with enhanced vertical mixing and lateral transport associated with strong wind forcing. Such conditions can induce internal waves, and shear at density interfaces, facilitating the redistribution of fine particles from the metalimnion and nearshore zones across the basin. Because sediment traps efficiently capture these fine, recently resuspended particles, they may overrepresent the biogenic and carbonate fractions relative to the mineral-rich bulk sediments accumulating in deeper lake zones (Horppila and Nurminen 2005; Lane et al. 2019).

These storm-driven resuspension events are critically linked to nutrient cycling in lakes. For example, sediment resuspension substantially contributes to internal phosphorus loading (Evans 1994; Scavia et al. 2022). The episodic nature of resuspension aligns with findings from lakes such as Lake Ontario, Lake Erie and Lake Chao, where wind and hydraulic disturbances drive nutrient release and sediment flux variability (Rosa 1985; Lin et al. 2021; Luo et al. 2022; Wang et al. 2024). Such dynamic processes underscore the importance of integrating high-frequency physical and biogeochemical observation to accurately capture sediment budget and nutrient dynamics.

Taken together, these events underscore how the interplay of extreme precipitation, soil moisture status, snow and glacier melt, and wind-driven resuspension shapes the Rhône–Lake Geneva system’s response to hydroclimatic extremes. Floods, droughts, and storms can produce fundamentally different sedimentation responses depending on hydrological conditions and stratification, through changes in river discharge, intrusion structure, and resuspension processes. More broadly, these mechanisms are characteristic of stratified, river-influenced lakes, highlighting the importance of high-frequency observations to resolve event-scale variability and improve predictions of sediment transport and biogeochemical responses under changing hydroclimatic conditions (Wilhelm et al. 2022).

Conclusions

This 5-year high-frequency study shows that the Rhône River is the primary driver of sedimentation in Lake Geneva, controlling both seasonal particle fluxes and responses to extreme events. Sediment-trap observations reveal that density-driven intrusions transport particles below the thermocline, sustaining high fluxes at mid-depth and in deep waters. While primary production and carbonate precipitation contribute to particulate fluxes, their influence remains secondary to fluvial inputs.

Our results highlight how contrasting extreme events modify sediment transport pathways. Floods enhance

sediment delivery and can shift transport between inter-flows and underflows, whereas storms and wind-driven events promote resuspension and redistribution of previously deposited material. These findings demonstrate that sedimentation dynamics depend not only on the magnitude of external forcing but also on its interaction with stratification and circulation.

This study also emphasizes the value of high-frequency in-lake observations. Unlike remote sensing approaches, which primarily capture surface SPM patterns, sediment traps and in situ measurements resolve vertical fluxes, subsurface transport, and in-lake processes. This process-based perspective enables a clearer separation of fluvial, biogenic, and carbonate contributions.

Overall, our results support a general framework for large, river-influenced lakes in which fluvial inputs dominate sediment supply, while internal processes modulate particle composition and vertical redistribution. The mechanisms identified here are likely applicable to other stratified lakes and highlight the need for high-resolution monitoring to improve predictions of sediment dynamics and biogeochemical responses under increasing hydroclimatic variability.

Given projections for more frequent and intense floods, droughts, and storms in alpine catchments, understanding the combined effects of discharge variability and extreme weather will be critical. This knowledge is essential for predicting how river–lake mixing, and hydrological extremes will shape future sedimentation patterns, water quality, and biogeochemical cycling in peri-alpine lakes. Long-term, high-resolution monitoring coupled with integrated modelling will remain essential for supporting sound management and adaptation strategies in these sensitive ecosystems.

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Author contributions E.F. and N.P. contributed to the conceptualization of the study, with N.P. leading the writing and manuscript revision. E.F. prepared most of the figures, while N.P. prepared Fig. 3. C.G. and J.K. managed the collection, lyophilisation, and weighing of sediment trap material. N.D. and D.J.J. supervised the chemical analyses performed by laboratory technicians and reviewed the manuscript critically. All authors reviewed and approved the final manuscript and agree to be accountable for their respective contributions.

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Data availability The sediment trap data are published openly on zenodo and can be accessed on <https://zenodo.org/records/17607859>.

Declarations

Conflict of interest The authors declare no competing interests.

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